

Array Description

Time limit: 1.00 s

Memory limit: 512 MB

You know that an array has n integers between 1 and m , and the absolute difference between two adjacent values is at most 1.

Given a description of the array where some values may be unknown, your task is to count the number of arrays that match the description.

Input

The first input line has two integers n and m : the array size and the upper bound for each value.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array. Value 0 denotes an unknown value.

Output

Print one integer: the number of arrays modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 100$
- $0 \leq x_i \leq m$

Example

Input	Output
3 5 2 0 2	3

Explanation: The arrays $[2, 1, 2]$, $[2, 2, 2]$ and $[2, 3, 2]$ match the description.